

Urate to CAD_UKB: MRMU vs MRAPSS Comparison

Generated with MRMU project workflow

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Purpose

This report adds a single-exposure MRAPSS result as a comparison to the MRMU analysis.

Both analyses use:

- Exposure: Biomarker_Urate
- Outcome: CAD_UKB
- IV threshold: 5e-5
- EUR LDSC scores
- local 1000 Genomes EUR PLINK clumping

The difference is:

method	confounders included
MRMU	Metabolic_SBP, Metabolic_DBP
MRAPSS	none

Results

method	beta	SE	p-value	IVs	effective IVs	pi0
MRMU(SBP+DBP con-founders)	0.012273	0.01526020	0.41919	696	322.2309	0.4629754
MRAPSS(0.04490893 con-founders)	0.01613808	5.3893e-03		696	324.7741	0.4666295

Interpretation

The single-exposure MRAPSS result, which does not adjust for SBP and DBP, estimates a larger urate effect on CAD_UKB and is statistically significant.

After adding SBP and DBP as confounders in MRMU, the estimated urate effect is smaller and no longer statistically significant.

This comparison suggests that part of the apparent urate-CAD_UKB association in the single-exposure analysis may be explained by shared genetic components with blood pressure traits.

Reproducibility

The MRAPSS comparison was run with:

```
MRMU_EXAMPLE_DIR=/path/to/MRMU_R_github Rscript scripts/run_urate_cad_mrapss_comparison.R
```

Key output files:

file	description
results/urate_cad_mrapss_comparison/MRAPSS_urate_mrapss_result.csv	MRAPSS_urate_mrapss result
results/urate_cad_mrapss_comparison/urate_cad_comparison.csv	urate_cad comparison
results/urate_cad_mrapss_comparison/urate_cad_mrapss_fit.rds	urate_cad_mrapss fit
results/urate_cad_mrapss_comparison/urate_cad_mrapss_inputs	urate_cad_mrapss inputs
results/urate_cad_mrapss_comparison/DSG_logging_and_model_fitting_log	DSG, logging, and model fitting log